

Template Practices Document

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Evaluated criteria

C1 Example criterion description.

C2 Example criterion description.

C3 Example criterion description.

1 Worked Solution Examples

Problems:

1.1. Example Uniform Investment Plan (C1 C2)

1.2. Example Linear Density Model (C2)

1.3. Example Table and Probability Summary (C3)

1.1 Example Uniform Investment Plan (C1 C2)

A variable V follows a uniform model on $[8, 16]$. Use the graph and density model to answer the tasks.

Questions/Tasks



C1 Write the density function and support.

C2 Compute $P(10 \leq V \leq 14)$ and justify with area.

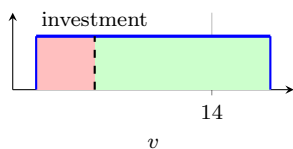
C1 - Write the density function and support

$$f_V(v) = \begin{cases} \frac{1}{16-8} = \frac{1}{8}, & 8 \leq v \leq 16, \\ 0, & \text{otherwise.} \end{cases}$$

$$8 \leq v \leq 16 \quad \longrightarrow \quad f_V(v) = \frac{1}{8} \quad \longrightarrow \quad \int_8^{16} f_V(v) dv = 1$$



C2 - Compute $P(10 \leq V \leq 14)$ and justify with area.



$$P(10 \leq V \leq 14) = \int_{10}^{14} \frac{1}{8} dv = \frac{1}{8}(14 - 10) = \frac{4}{8} = 0.5$$

The shaded area represents half of the support interval.



1.2 Example Linear Density Model (C2)

Suppose X has density $f(x) = kx$ for $0 \leq x \leq 2$.

Questions/Tasks

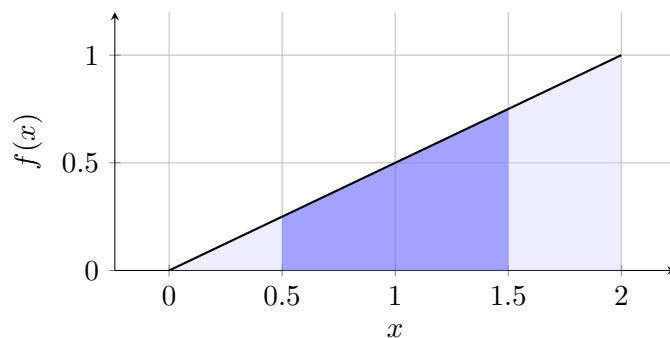


C2.1 Find the value of k .

C2.2 Compute $P(0.5 \leq X \leq 1.5)$.

C2.1 - Find the value of k .

$$1 = \int_0^2 kx \, dx = k \left[\frac{x^2}{2} \right]_0^2 = 2k \Rightarrow k = \frac{1}{2}$$



C2.2 - Compute $P(0.5 \leq X \leq 1.5)$.

$$P(0.5 \leq X \leq 1.5) = \int_{0.5}^{1.5} \frac{1}{2}x \, dx = \left[\frac{x^2}{4} \right]_{0.5}^{1.5} = \frac{1.5^2 - 0.5^2}{4} = 0.5$$



1.3 Example Table and Probability Summary (C3)

A summary table is shown for two intervals of a piecewise density.

Questions/Tasks

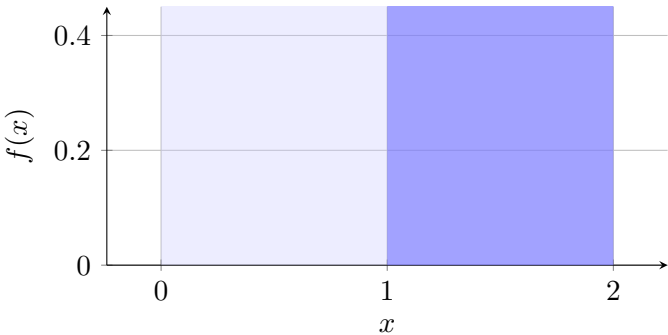


C3.1 Complete the probability summary table.

C3.2 State one consistency check.

C3.1 - Complete the probability summary table.

Interval	Expression	Result
$[0, 1]$	$\int_0^1 0.4 \, dx$	0.4
$[1, 2]$	$\int_1^2 0.6 \, dx$	0.6
Total	$0.4 + 0.6$	1



C3.2 - State one consistency check.

A valid check is that all interval probabilities are nonnegative and add to 1.



2 Graph Command Examples

This section provides one representative command from each graph file in book/preamble/graphs.

Problems:
2.1. Example from <code>uniform_linear_probability_distributions.tex</code>
2.2. Example from <code>probability_distribution_graphs.tex</code>
2.3. Example from <code>normal_distribution_graphs.tex</code>

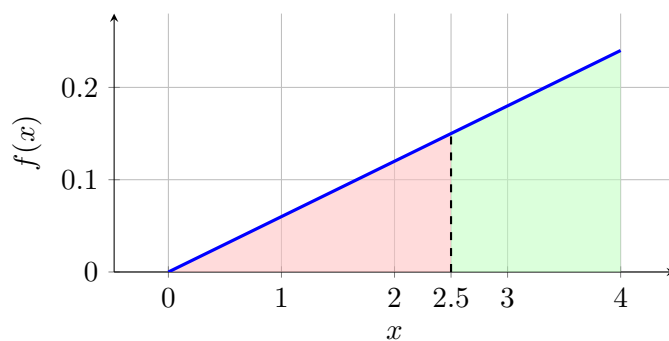
2.1 Example from `uniform_linear_probability_distributions.tex`.

Questions/Tasks



C2 Plot the linear density and shade the region below the threshold.

C2 - Plot the linear density and shade the region below the threshold.



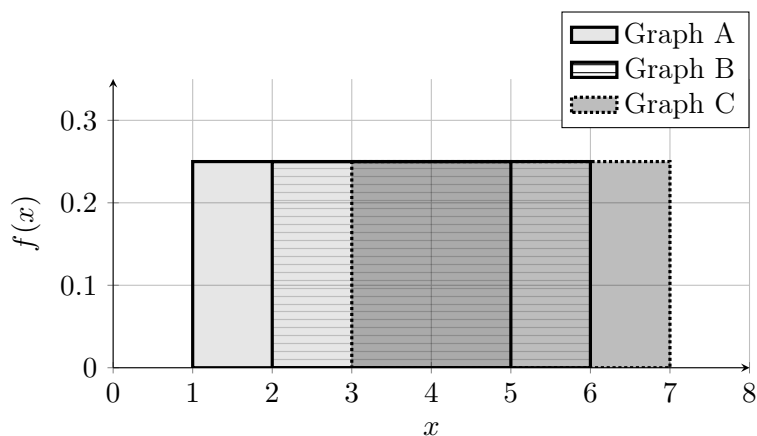
2.2 Example from probability_distribution_graphs.tex.

Questions/Tasks



C2 Plot the uniform density with three shaded intervals.

C2 - Plot the uniform density with three shaded intervals.



2.3 Example from normal_distribution_graphs.tex.

Questions/Tasks



C2 Plot the standard normal density and shade $P(Z \leq -0.75)$.

C2 - Plot the standard normal density and shade $P(Z \leq -0.75)$.

